

```
creating dataset
import pandas as pd
```

```
movies = [ (here are the movies but I cant upload the entire set)]
df = pd.DataFrame(movies)
df.to_csv("movies.csv", index=False)
print(f"Created movies.csv with {len(df)} movies.")
print(df[["title", "vote_average", "release_date", "runtime"]].head())
```

```
setting up
import streamlit as st
import pandas as pd
import ast
from pathlib import Path
```

```
st.set_page_config(page_title="Movie Recommender", layout="wide")
```

```
DATA_PATH = Path(__file__).parent / "movies.csv"
```

```
cleaning the dataset
```

```
@st.cache_data
```

```
def load_data():
```

```
    df = pd.read_csv(DATA_PATH)
```

```
def extract_genres(genre_str):
```

```
    try:
```

```
        genres = ast.literal_eval(genre_str)
```

```
        return [g["name"] for g in genres]
```

```
    except Exception:
```

```
        return []
```

```
df["genre_list"] = df["genres"].apply(extract_genres)
```

```
df["primary_genre"] = df["genre_list"].apply(lambda g: g[0] if g else "Unknown")
```

```
df["year"] = pd.to_datetime(df["release_date"], errors="coerce").dt.year
```

```
df = df.dropna(subset=["title", "vote_average", "year", "runtime"])
```

```
df = df.drop_duplicates(subset=["title"])
```

```
df = df.reset_index(drop=True)
```

```
return df
```